Introduction

Aphasia resulting from brain damage creates severe communication barriers for patients and their caregivers.

uh uh it's kiss kiss uh um hug hug uh and uh actually she was uh married uh uh æbɹɪl . uh uh United States uh no Coast card . and she uh keeps me uh jet uh oh god I don't know how to say it .

Text Only Large Language Model (LLM) :

Do you mean "It's about kissing and hugging. She was actually married, and she was from the United States, serving in the Coast Guard. And she sustains me; I just can't find the words to express it." ?

Common AAC System:

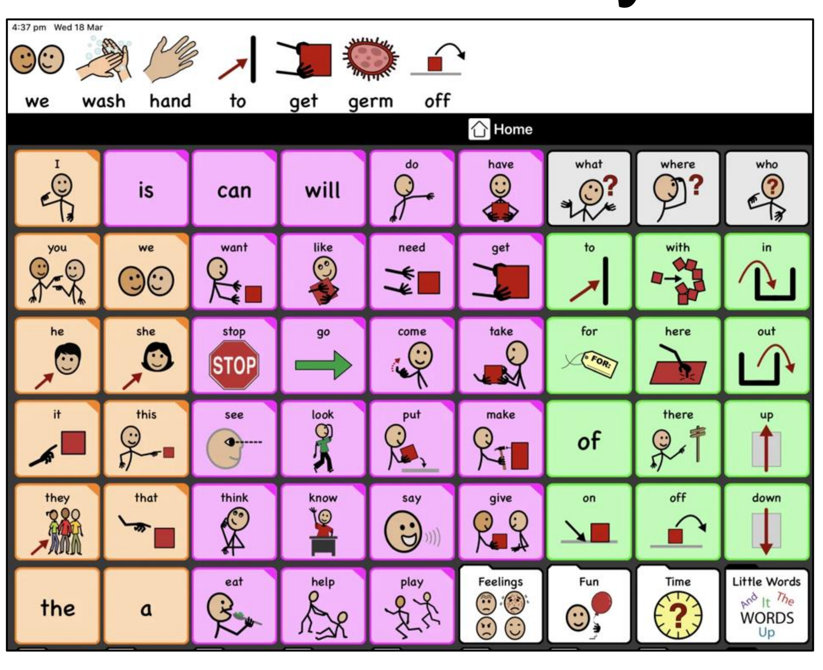
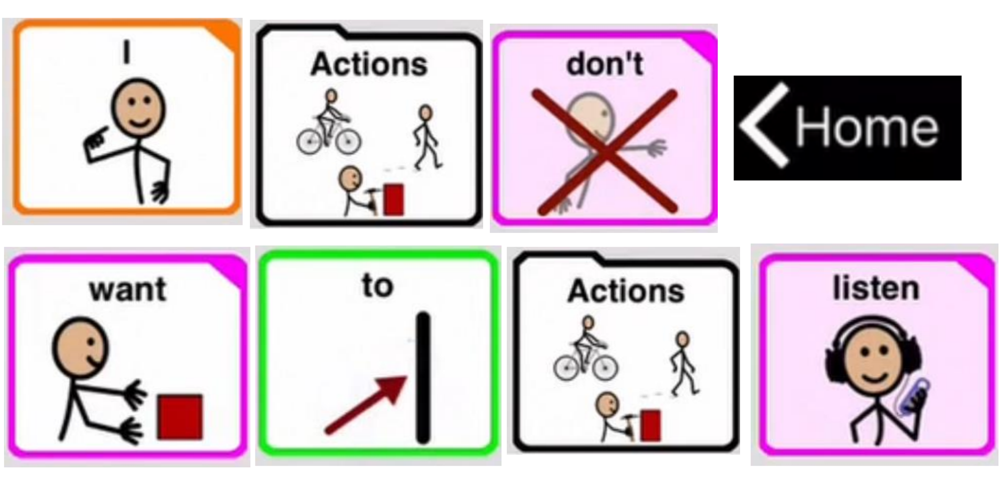


Figure 1: Proloquo2Go



Multiple interpretation:

- The noise is too loud, turn it off.
- I don't like this music, change one.
- I refuse listening to your advice.

From *Proloquo2Go with a message about washing germs off for the coronavirus* [Image], by AssistiveWare, 2020, AssistiveWare (https://www.assistiveware.com/blog/using-proloquo2go-to-talk-about-the-coronavirus).

- Sequential symbol selection imposes a high cognitive load and is too slow for natural conversation.
- Hard to cover diverse real-world scenarios.

Objective

To enhance the communication fluency and accuracy of assistive technologies for individuals with aphasia.

Background

Types of Aphasia

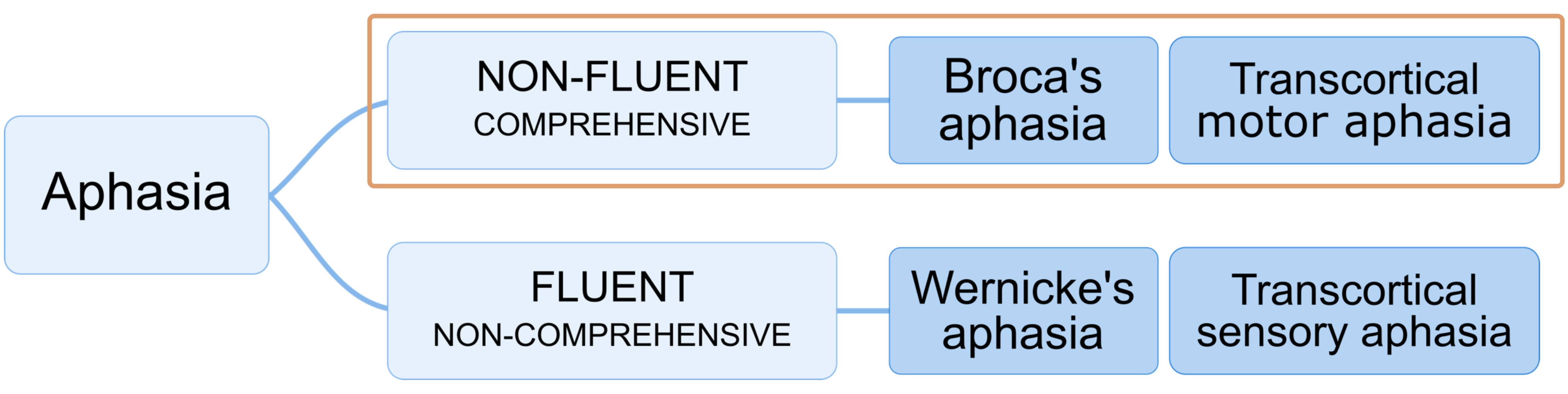


Figure 2: Classification of Aphasia

(Chart created by Finalist using Draw.io, 2025; Adapted from National Aphasia Association [1])

Recent Work

Table 1: Comparison of recent studies on aphasia

	Van Vaals et al. (2025) [2]	Shi et al. (2025) [3]	Manir et al. (2024) [4]
Dataset	Clinical Dataset + POS-Rule Synthetic data	LLM Simulation with Prompt Engineering	AphasiaBank database transcripts.
Restore Method	Finetune T5 & FLAN T5 with Synthetic data	ASR-LLM	Fine-tuned BERT (Mask Prediction)
Modality	Transcripts		
Results	Cos. Sim. : ~0.88 Qualitative evaluation Overlooks raw ASR errors.	BERTscore:~ 0.88 BLEURT: ~ -0.6 Limited generation quality	Human Rating : ~ 3/5 Fleiss' Kappa: 0.32 Moderate rating, low agreement.
Limita-tions	• Human-annotated transcripts may overlook the phonetic errors in raw ASR. • Synthetic dataset may lack clinical complexity. • Single-modality focus may miss key clues, hindering aphasic sentence reconstruction.		

(Data Table created by Finalist using PowerPoint, 2025)

Proposed Work

Speech Representation

- Acoustic Voice
- Lip-Reading Cues

Contextual Information

- Environment
- Conversation

AphaSpeak

Expressive Behavior

- Postural Movements
- Facial Expressions

Personalized Features

- Patient habits
- Pronunciation errors

- Incorporate multimodal inputs
- Built on raw ASR transcripts from clinical dataset.
- Optimized for real-time clinical deployment.

System Design

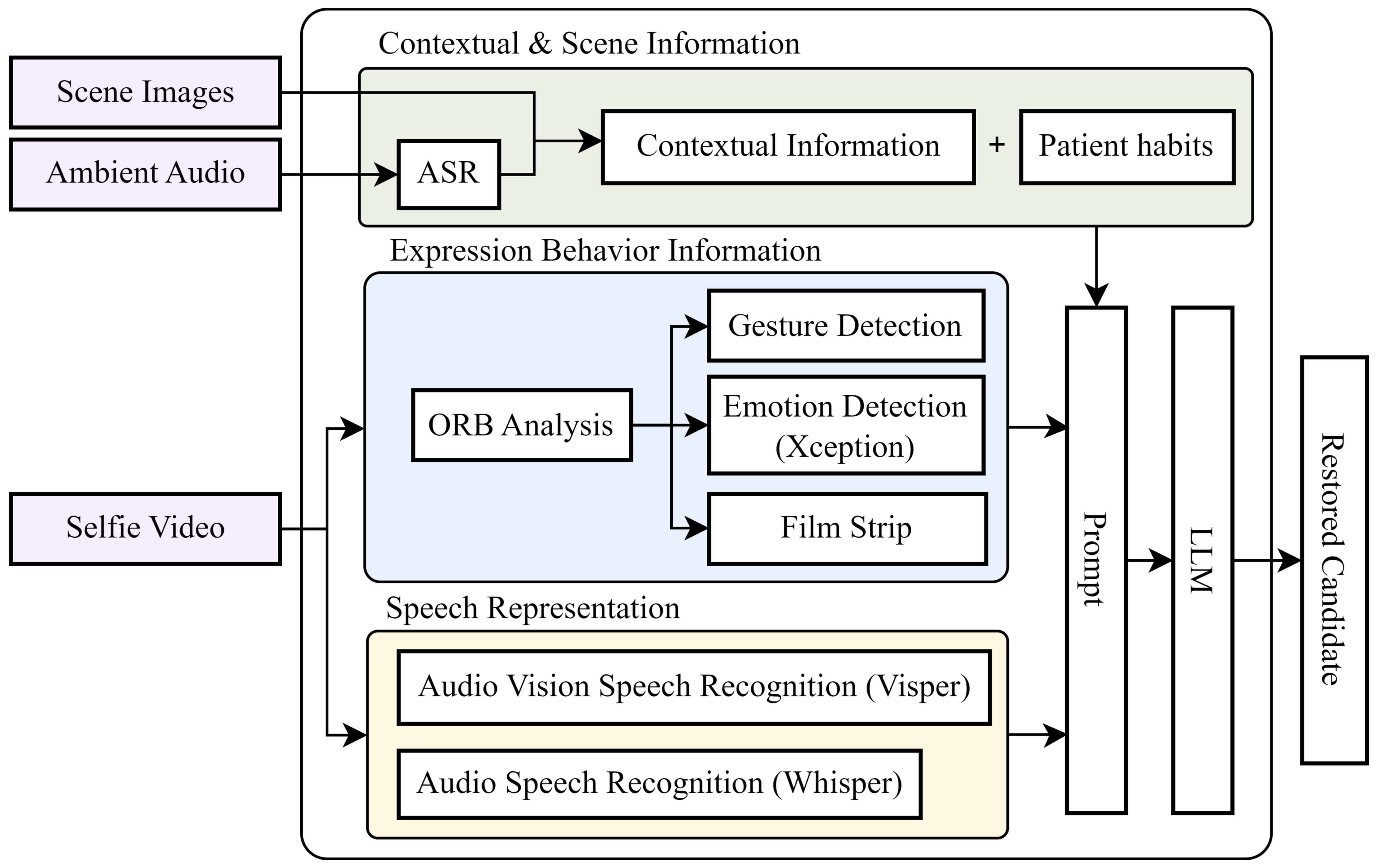


Figure 3: System Design (Chart created by Finalist using Draw.io, 2025)

Contextual & Scene Information

- Transfer ambient audio into **transcripts** to build contextual information.
- Scene images are provided as primary context to interpret the **conversation topic** .
- Patient habits summarized from history support pronunciation correction.

Expression Behavior Information

- ORB extracts **keyframes** to reduce compute [5].
- Gestures and facial expressions help infer **intonation and intent** [6][7].
- Keyframes are combined into a filmstrip to capture additional **actions or references**.

Speech Representation

- Visper performs audio-visual speech recognition using **lip movements** to improve recognition when speech articulation is unclear [8].
- Whisper and Visper enables **cross-modal verification** to resolve ambiguous phonemes [9].

Experimental Results

Dataset and Evaluation Metrics

- Broca's and Transcortical motor aphasia from AphasiaBank dataset [10]
- Adapted / Merged : 4520 sentences / 1128 sentences
- Evaluation Metrics : BLERUT / BLERUT-20 / BERT-Cosine Similarity

Output Quality

- Measure the output accuracy across Top-1 to Top-5 candidate suggestions.
- At Cos.Sim. ≥ 0.85, **Top-3 accuracy exhibits a marked improvement** over Top-1 (from 64% to 80%) and closely approaches the Top-5 accuracy of 84%.
- AphaSpeak consistently **outperforms the text-only LLM**, with a 10% margin.

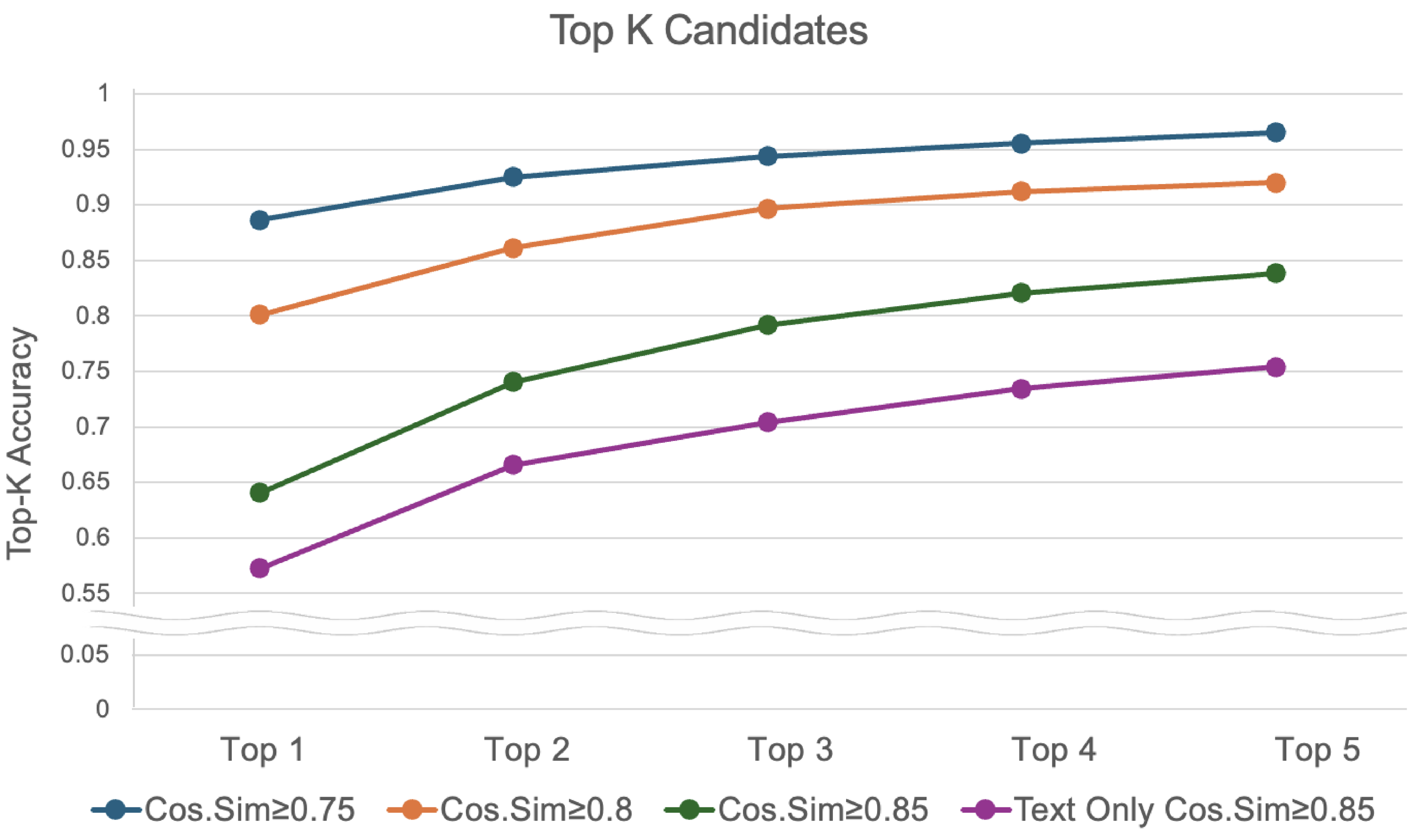


Figure 4: Output Quality Experiment Result
(Graph created by Finalist using Excel, 2025)

Ablation Study

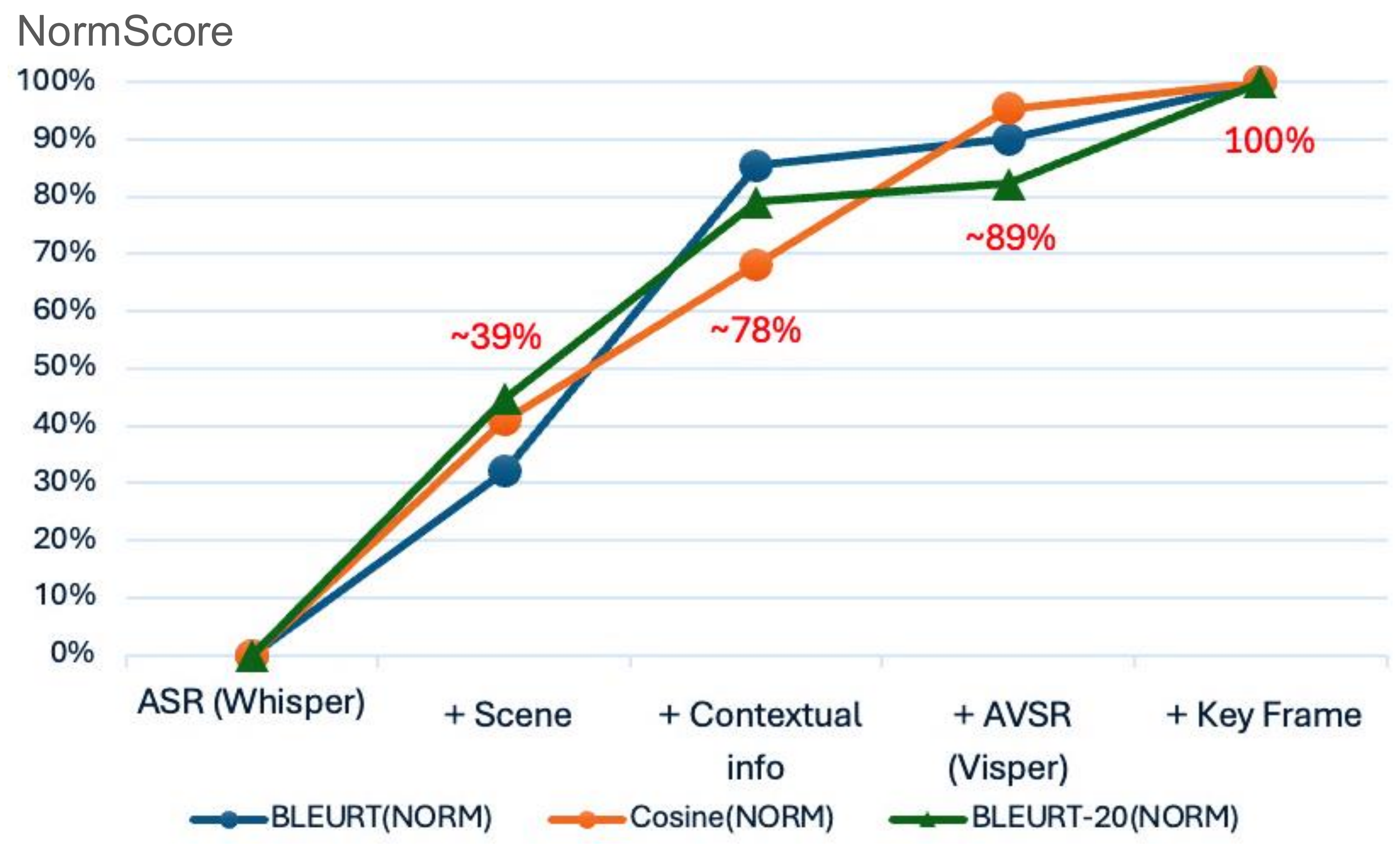


Figure 5: Ablation Experiment Result
(Graph created by Finalist using Excel, 2025)

$$NormScore = \frac{X - X_{min}}{X_{max} - X_{min}}$$

X: performance of a modality configuration
X_{min}: baseline model
X_{max}: full multimodal system

- **Both Contextual and Scene Information** contribute to around **30-40% improvement**.
- **Visper and Expression Behavior Information** provide a **25% overall improvement**.

Optimization Experiment

- **Concise COT & Semantic Fusion :** Fuse multimodal features with concise COT architecture.
- **Asynchronous Streaming :** Enables continuous streaming via WebSockets and async queues.
- **Priority-Aware Resource Scheduling** : Optimizes resource allocation via priority-based task management.

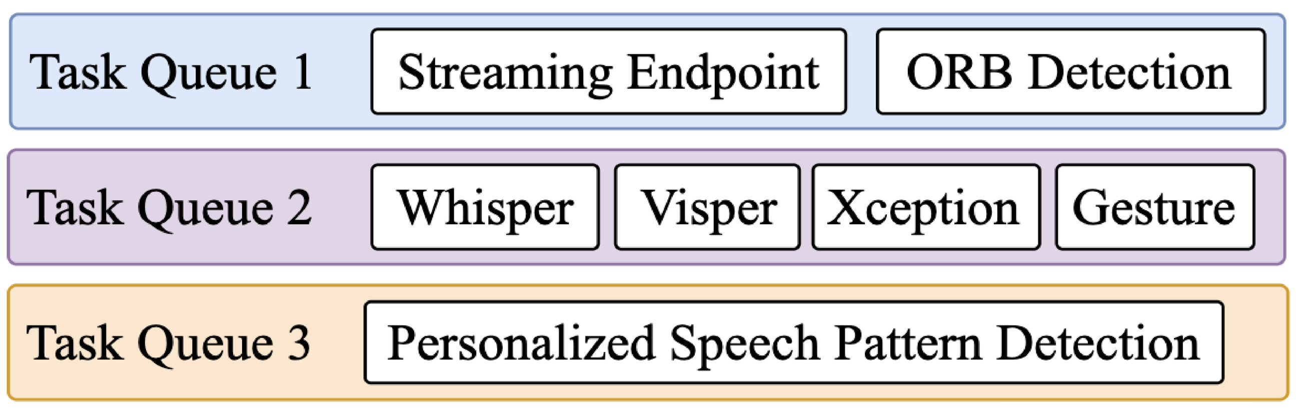


Figure 6: Server Task Queuing
(Graphic created by Finalist using Draw.io, 2025)

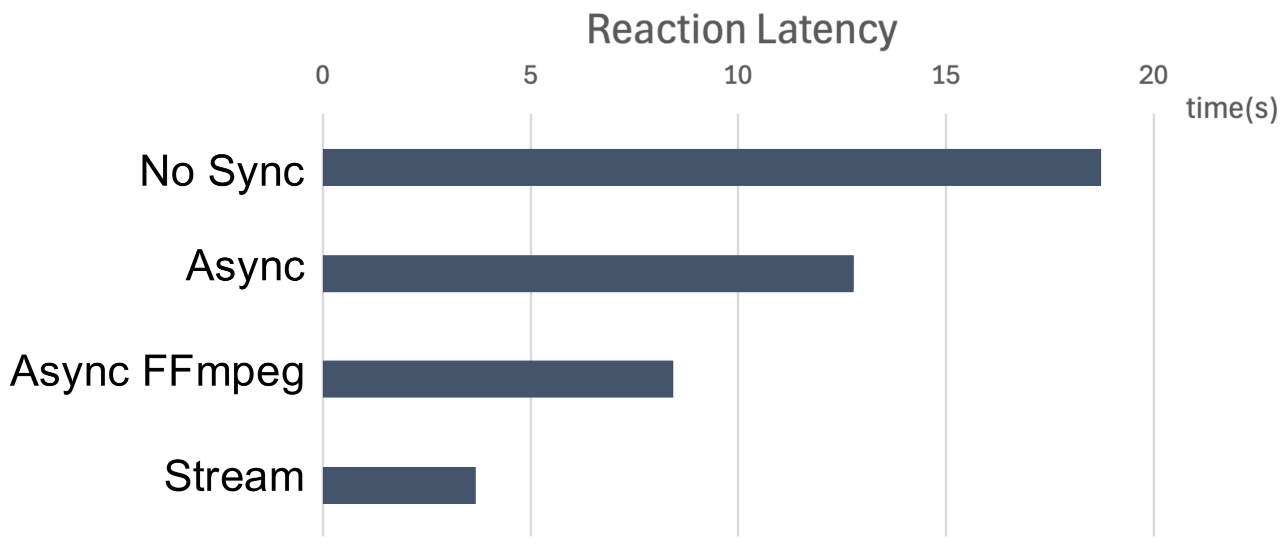


Figure 7: System Architecture Diagram
(Graph created by Finalist using Excel, 2025)

Practical Considerations

- Distilled multimodal reasoning from Gemini 2.5 Flash into a compact 14B model.
- Through **Choice-Filtered Distillation**, AphaSpeak **surpasses Gemini (+0.04 BLEURT-20 & +0.04 Cos.Sim.).**

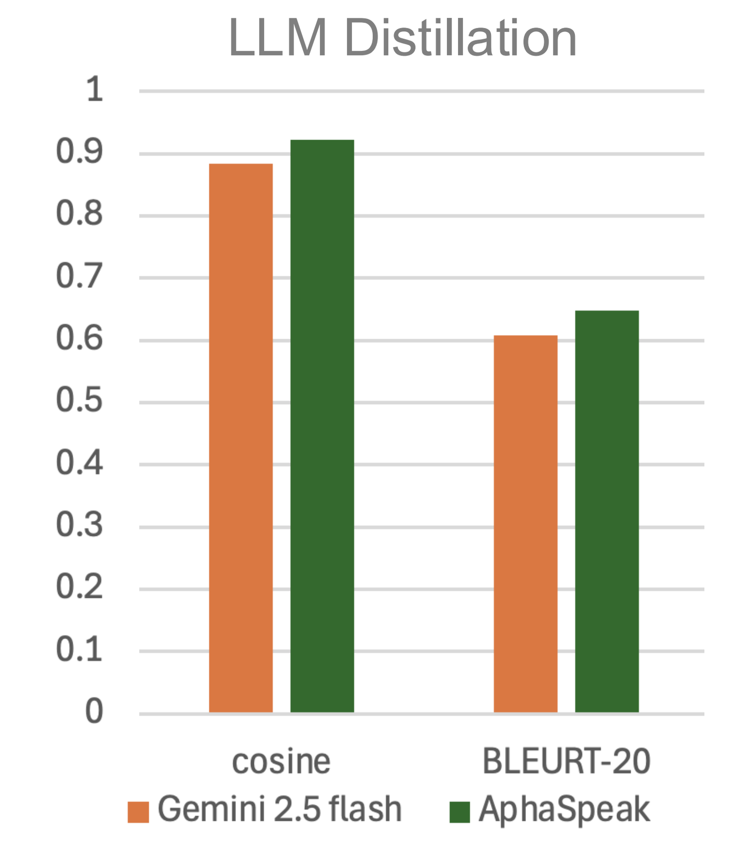


Figure 8: LLM Distillation
(Graph created by Finalist using Excel, 2025)

System Validation

- AphaSpeak architecture is extensible to various languages, including English and Chinese.

Table 2: System Validation on English and Chinese Dataset

Metrics	English	Chinese
BLEURT-20	0.6129	0.4300
BLEURT	–0.067	
Cosine Similarity	0.9134	0.8299

(Data Table created by Finalist using PowerPoint, 2025)

Examples

Example 1

Interviewer :

"thinking back can you tell me a story about something important that happened to you in your life ?"

Patient :

"um words uh this this pɹɛʃɪn of a group um many many years ago . uh I don't know that sometimes it's easy . w words and sentences."

Gemini 2.5 Flash (Text Only): "Um, this group formed many, many years ago. Uh, I don't know, sometimes it's easy to forget."

AphaSpeak (Ambient Audio+Selfie Video): "Many years ago, I found it difficult to express myself in words and full sentences, particularly when under group pressure."

Example 2

(From *Cat* [Image], by AphasiaBank, n.d., TalkBank. https://talkbank.org/aphasia/)

Interviewer :

"Take a look and see if you can tell me a story about what's going on in that picture ."

Figure 9: Visual data samples

Patient :

"and then she has a little the she's got to have that . and then this [point dad] grandma [point fireman] or g uh grandma she has that . and then then grandma no [wave hands] it's that's not right . um and she hadta go over here again . that's uh mm ...... and grandma ha uh moth no . fath father father yeah ."

Gemini 2.5 Flash (Text Only): "She had to go and see her father."

AphaSpeak (Multimodal): "The girl wants the cat, and her father is in the tree trying to help. The firefighters are bringing a ladder because the father needs to come down."

Conclusions

AphaSpeak enables fluent, real-time communication for individuals with aphasia.

- **Enhanced Accuracy**: Integrates visual and contextual data to fix raw ASR errors, restoring semantic depth and raising BLEURT-20 to 0.61 and Cosine Similarity to 0.91.
- **Optimized Deployment**: Through architectural tuning and choice-filtered distillation, system achieves low-cost, high-efficiency deployment with computational latency under 4 seconds.
- **Empowering Natural Communication**: Simplifies word-finding through a simple selection interface to eliminate conversational barriers, allowing for effortless and meaningful participation in daily life and social interactions.

References

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[10] MacWhinney, Fromm, Forbes, Holland. AphasiaBank: Methods for studying discourse. *Aphasiology*, 2011.